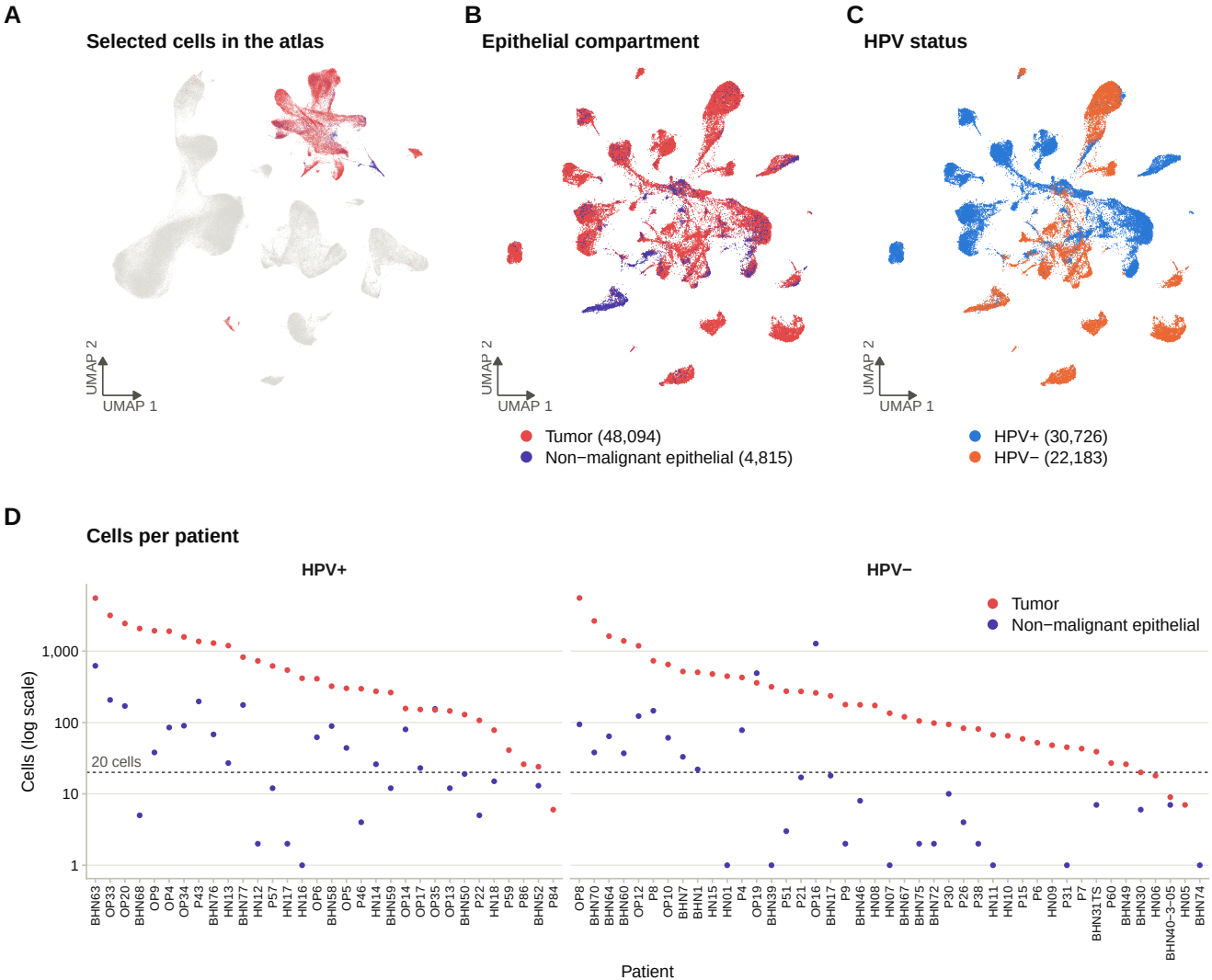

A
Selected cells in the atlas
UMAP 2
UMAP 1
B
Epithelial compartment
UMAP 2
UMAP 1
Tumor (48,094)
Non−malignant epithelial (4,815)
C
HPV status
UMAP 2
UMAP 1
HPV+ (30,726)
HPV− (22,183)
D
Cells per patient
HPV+
HPV−
Tumor
Non−malignant epithelial
Cells (log scale)
1,000
100
10
1
20 cells
BHN63 OP33 OP20 BHN68 OP9 OP4 OP34 P43 BHN76 HN13 BHN77 HN12 P57 HN17 HN16 OP6 BHN58 OP5 P46 HN14 BHN59 OP14 OP17 OP35 OP13 BHN50 P22 HN18 P59 P86 BHN52 P84
OP8 BHN70 BHN64 BHN60 OP12 P8 OP10 BHN7 BHN1 HN15 HN01 P4 OP19 BHN39 P51 P21 OP16 BHN17 P9 BHN46 HN08 HN07 BHN67 BHN75 BHN72 P30 P26 P38 HN11 HN10 P15 P6 HN09 P31 P7 BHN31TS P60 BHN49 BHN30 HN06 BHN40−3−05 HN05 BHN74
Patient